

Batch: DMW 2 Roll No.: 16010324044
Experiment/assignment/tutorial No.: 06
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Ensemble learning using Bagging, Boosting, Random Forest and Stacking

AIM: a) To learn application of Bagging, AdaBoost, Random Forest and Stacking techniques to improve performance of baseline machine learning algorithm

OUTCOME: Students will be able to apply the ensemble techniques to improve classification performance of weak learner for efficient Data mining

Procedure:

Carefully observe the arguments and their default values for the following sklearn functions, by visiting the respective websites

- a) sklearn
AdaBoostClassifier(<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html>)
- b) sklearn BaggingClassifier
(<https://scikit-learn.org/0.16/modules/generated/sklearn.ensemble.BaggingClassifier.html>)
- c) sklearn RandomForestClassifier
(<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>)
- d) sklearn StackingClassifier
(<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.StackingClassifier.html>)
- e) For dataset description, visit link,
<https://archive.ics.uci.edu/ml/datasets/Raisin+Dataset>

1. Read the raisin data set, Raisin.csv, collected for binary classification of raisin varieties Kecimen and Besni, using features derived from raisin images. Describe the dataset briefly
2. Encode response variable “Class” into categorical variables; Note dataset variable types; Shuffle dataset; Separate predictor and response variables into different data frames; Split the data into train and test data sets and find size of each
3. Perform classification on train and test data using full grown Decision Tree classifier trained on train data. Comment on the classification results obtained on train and test data

4. Perform grid search cross validation using train data to find hyper-parameters of the decision tree model. Train decision tree model using best hyper-parameters. Evaluate performance of the tuned model on train and test data. Comment on the results obtained.
5. Perform classification on train and test data using **Bagging classifier** with default parameter values. Evaluate performance of the trained model on train and test data. Comment on the results obtained. .
6. Perform classification on train and test data using **AdaBoost classifier** with default parameter values. Evaluate performance of the trained model on train and test data. Comment on the results obtained.
7. Perform classification on train and test data using **Random Forest classifier** with default parameter values. Evaluate performance of the trained model on train and test data. Comment on the results obtained.
8. Perform **stacking** with k-nn and Pruned DT as base learners and Logistic Regression classifier as meta learner; report your observations and results

Do not perform GridSearchCV for Bagging, Boosting, Random Forest and Stacking

Observation Table:

Model	Train Accuracy (%)	Test Accuracy (%)	Train-Test Gap (%)	F1-Score (%)	Overfitting?
Tuned Decision Tree	87	86	1	78	No
Bagging	99	84	15	84	Yes
AdaBoost	88	87	1	87	No
Random Forest	100	86	14	86	Yes
Stacking	91	85	6	85	No

Train-Test Gap = Train accuracy – Test accuracy

Analyze:

1. Which model gives the **highest test accuracy**?
AdaBoost

2. Which model gives the **highest F1-score**?

3. Which model shows the **largest difference between training and testing performance**?

Tuned Decision Tree

4. Which ensemble technique improves upon the baseline Decision Tree?

Tuned Decision Tree

5. Does an ensemble always perform better than a single Decision Tree? Justify from your results.

Tuned Decision Tree Test Accuracy

An ensemble does not necessarily outperform a single Decision Tree for every dataset. Your justification should be based on the actual test accuracies and F1-scores in your observation table.

6. Which ensemble method would you recommend for this dataset and why?

AdaBoost, since it does not overfit with the dataset

7. Which model has the largest Train–Test Accuracy Gap? Does this indicate overfitting? Compare the gap across all five models and justify your answer.

Bagging, A large gap means the model performs considerably better on training data than unseen test data, which is an indication of overfitting.

Conclusion

In this experiment, I implemented ensemble techniques like Bagging, AdaBoost, Random Forest and Stacking on the Raisin dataset. Their classification performance was evaluated using accuracy and F1-score. The results showed that ensemble methods can improve generalization compared with a single Decision Tree, with AdaBoost achieving the highest test accuracy among the obtained results.

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